

CyberTutor: AI-Powered Security Agent

Integrating Local LLMs with Live Threat Intelligence

Executive Summary

- **The Problem:** Cybersecurity analysts face information overload and must constantly switch between different databases (NVD, MITRE, Shodan) to investigate threats.
- **The Solution:** A centralized, AI-driven terminal agent that fetches, reads, and summarizes live intelligence on command.

System Architecture

- **Core Logic:** Python
- **The Brain:** Local Llama 3 (ensures zero data leakage).
- **The Interface:** `rich` library for advanced terminal UI.
- **The Tools:** REST APIs for automated data retrieval.

Core Feature 1 - Vulnerability Tracking

- **Real-Time CVE Intelligence**
- Demonstrates the ``cve [keyword]`` command.
- Pulls raw JSON data from the National Vulnerability Database and translates it into an actionable summary with mitigation steps.

Core Feature 2 - Threat Actor Tactics

- **MITRE ATT&CK Integration**
- Demonstrates the ``mitre [keyword]`` command.
- Helps junior analysts rapidly understand complex adversary techniques and TTPs (Tactics, Techniques, and Procedures).

Core Feature 3 - Infrastructure Reconnaissance

- **Shodan Host Profiling**
- Demonstrates the ``shodan [IP]`` command.
- Identifies open ports, organizations, and vulnerabilities for specific IP addresses on the fly.

UI & User Experience

- **Terminal Modernization**

- A side-by-side comparison of raw Python API output vs. the clean, panel-based `rich` interface.
- Security tools need to be readable to be effective during rapid incident response.

Automated Session Auditing

- **Verifiable AI Interactions**
- Automated text logging system integration.
- In an enterprise environment, all AI interactions and intelligence queries must be logged for auditing and compliance.

Technical Challenge - Context Windows

- **Managing AI Memory**
- Encountered a "hallucination" bug when the AI got overwhelmed by too much data.
- **Solution:** Used list slicing (``conversation[:1]``) to clear the AI's short-term memory before feeding it new API data.

Security & Privacy

- **Why Local LLMs Matter**

- Running Llama 3 locally ensures data sovereignty compared to cloud APIs like ChatGPT.
- Analyzing proprietary network data or internal IP addresses requires absolute privacy, which local models guarantee.

Project Impact

- **Efficiency and Speed**
- Reduces the time it takes to gather preliminary threat intelligence from minutes (manual searching) to seconds.
- Consolidates multiple browser tabs into a single terminal window.

Roadmap & Future Features

- **Next Steps**

- Adding VirusTotal API for malware hash lookups.
- Implementing a web-based dashboard interface.
- Expanding the AI to ingest and analyze local log files (like `.pcap`` or Syslog).